

EASY VISA

An optimization problem | 15-Dec-2025

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- Business Problem Overview and Solution Approach
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Executive Summary

- The goal of this project was to develop a robust machine learning classifier to predict the visa application outcomes and identify the drivers for approvals. By automating the initial screening process, the model significantly reduces the manual effort and enhances the operational efficiency.
- **Methodology & Development**
 - **Benchmarking:** A baseline is established with default models, followed by the development of advanced ensemble models for robust predictions.
 - **Data processing:** Implemented Class Imbalance mitigation to balance the output classes and eliminate bias towards under-represented class.
 - **Hyperparameter Tuning:** Applied optimization techniques to tune the model and maximize the performance.
- **Model Evaluation and Performance:**
 - The goal is to minimize False Positives (a wrong person getting the position) and False Negatives (Will miss a qualified resource)

Executive Summary

- **Model Evaluation and Performance (Continued):**

- **Gradient Boosting Classifier** achieved the best F1 Score of 0.82 indicating a balance between Recall (False Negatives / missed qualified resources) and Precision (False Positives / incorrect approvals).
- The model achieved a high Recall (~89%) on the positive class, successfully identifying vast majority of qualifying applications.
- ~26% of applications were predicted incorrectly, however the performance was consistent across the Training, Validation and Test data indicating that model was generalizing well.

- **Key Insights and Drivers**

- **Employee education, Prevailing wages and Job experience** are top features influencing the decision.
- Higher education, higher prevailing wages and prior job experience all show a strong positive correlation with approval ratings.
- Applications from Europe region show highest approval rates relative to their volume, while South America recorded the lowest approval rates.

Business Problem Overview and Solution Approach

Business Problem

The US Immigration and Nationality Act (INA) allows US employers to hire foreign workers on both temporary and permanent visa bases. To do so, the employers must file job certification applications seeking approvals from Office of Foreign Labor Certification (OLFC).

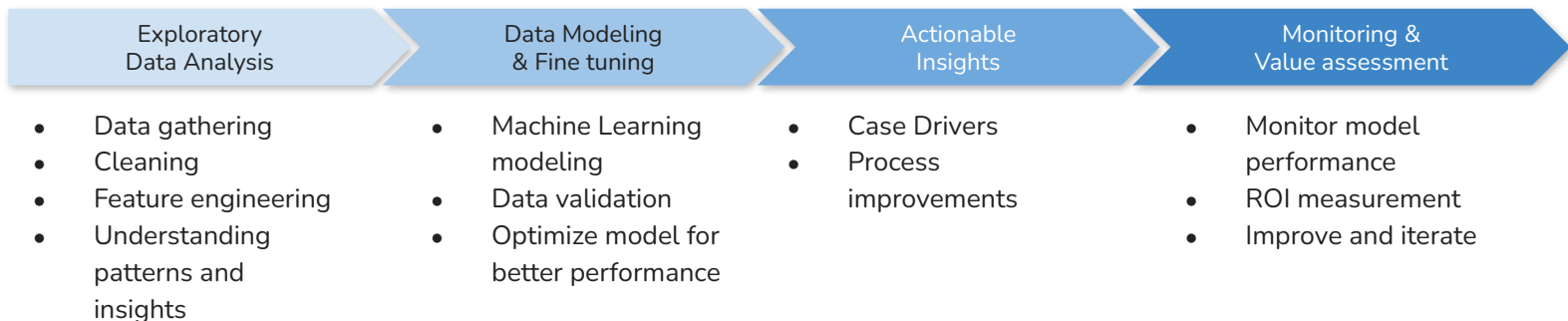
The volume of these job applications is increasing significantly year over year making the manual review increasingly ineffective and tedious.

Objective

To develop a data driven solution that streamlines the process by identifying the applications with higher chance of approvals.

The solution will enable OLFC to prioritize review efforts efficiently and significantly reduce manual processing time.

Solution Approach



Data Overview



Data Source
EasyVisa.csv

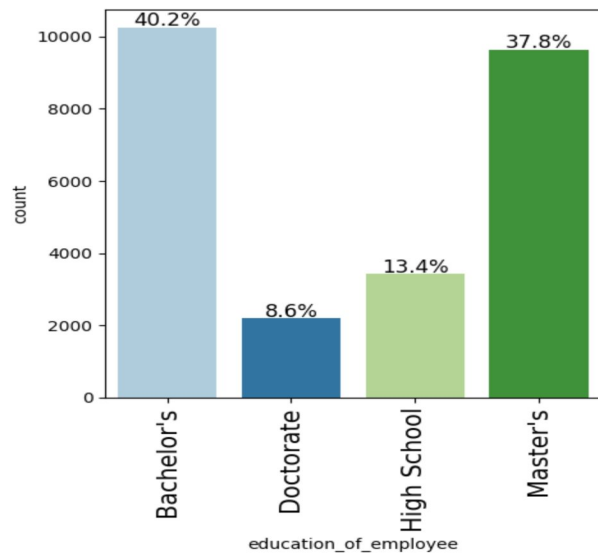
Online orders made by
Foodhub customers

Size of the Data
25,480 rows | 12 cols

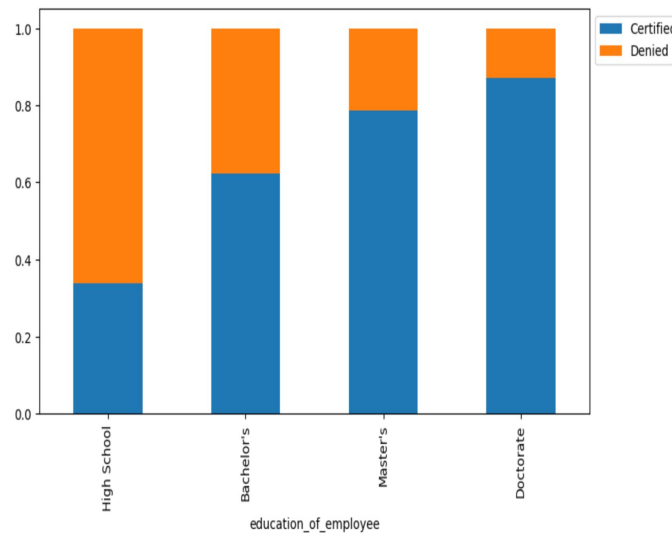
- 25,480 applications
- 5 regions of employment
- Applications from 5 continents and Oceania region
- 66.8% of certified applications.

	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage
0	Asia	High School	N	N	14513	2007	West	592.2029
1	Asia	Master's	Y	N	2412	2002	Northeast	83425.6500
2	Asia	Bachelor's	N	Y	44444	2008	West	122996.8600
3	Asia	Bachelor's	N	N	98	1897	West	83434.0300
4	Africa	Master's	Y	N	1082	2005	South	149907.3900
5	Asia	Master's	Y	N	2339	2012	South	78252.1400
6	Asia	Bachelor's	N	N	4985	1994	South	53635.3900
7	North America	Bachelor's	Y	N	3035	1924	West	418.2298
8	Asia	Bachelor's	N	N	4810	2012	Midwest	74362.1900
9	Europe	Doctorate	Y	N	2251	1995	South	67514.7600

EDA : Applicants Education

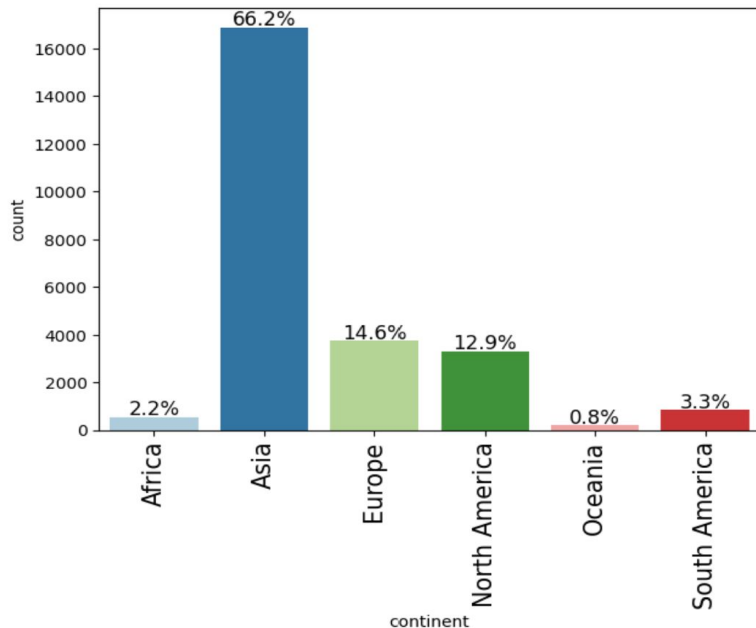


- Nearly 87% of applicants hold a Bachelor's degree or higher educational qualification.
- The remaining 13% of applicants possess a high school education

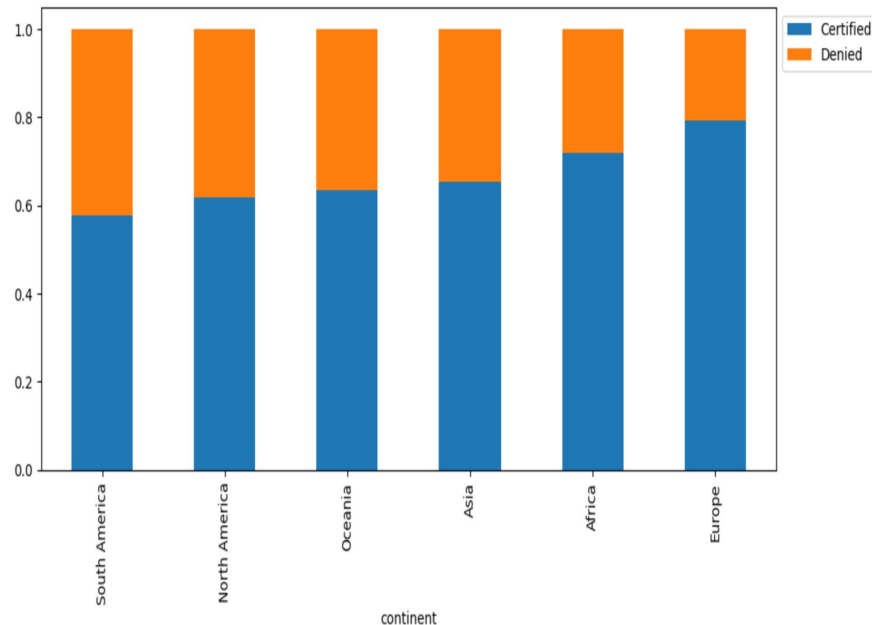


- Education level demonstrates a strong positive correlation with application approval rates.
- Specifically, 85% of doctorate-level applications and 80% of master's-level applications have received approvals.

EDA : Applicant's Continent

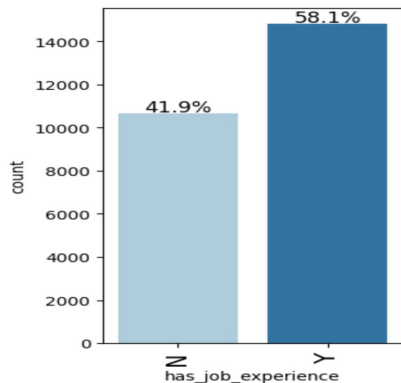


- A significant (66%) majority of the applications are from Asia.

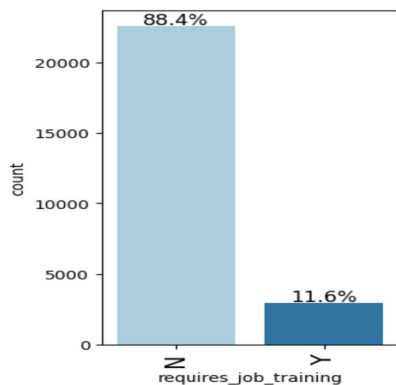
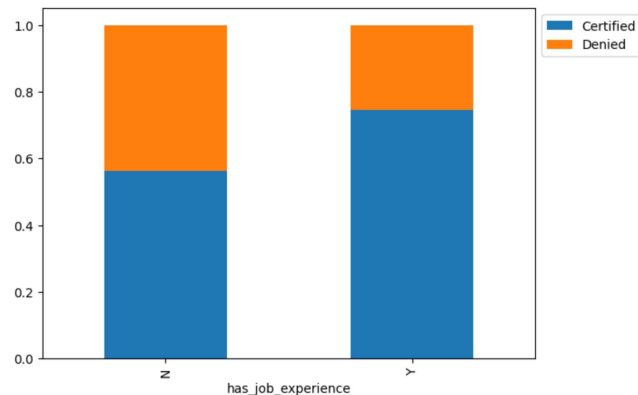


- Nearly 80% of applications from Europe are approved.
- South America records the lowest approval rate across all continents, at 58%.

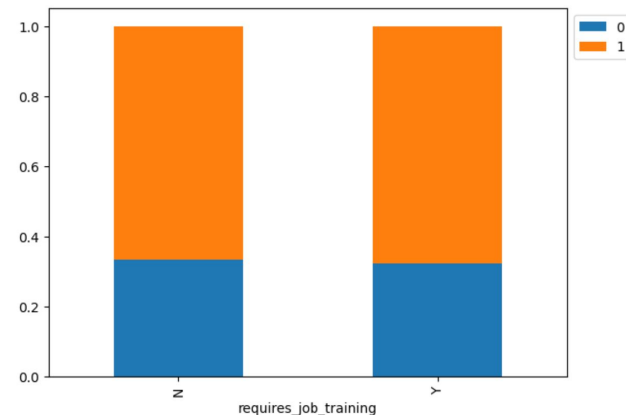
EDA : Applicants Experience & Training requirement



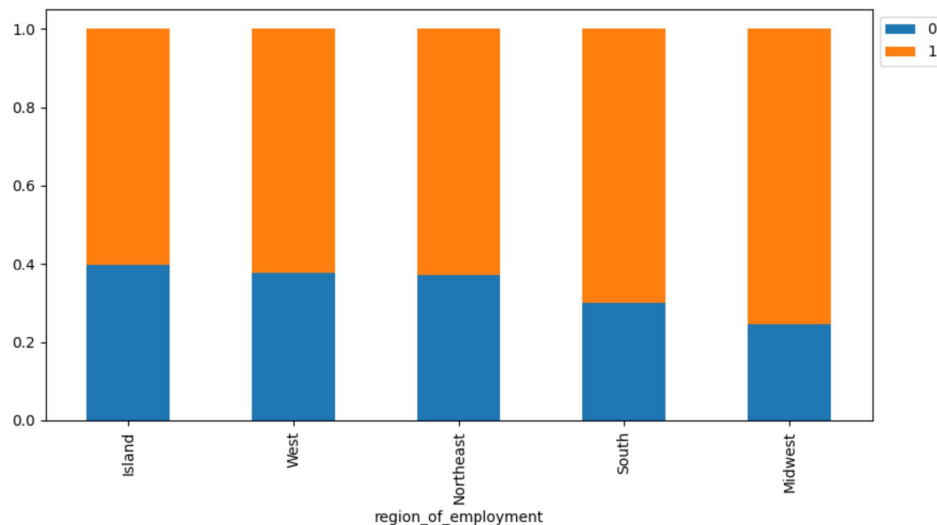
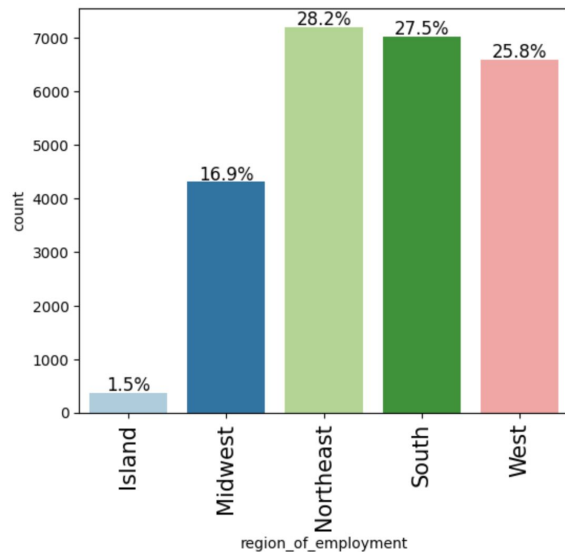
- Forty-two percent of candidates reported having no prior job experience.
- Applicants who do possess relevant job experience exhibit higher approval rates.



- Twelve percent of applicants indicate a requirement for job training.
- Approval rates remain similar for these applicants compared to those who do not require training.

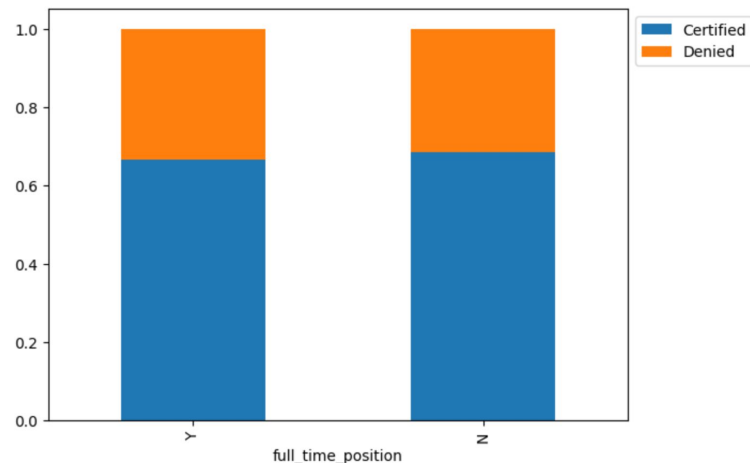


EDA : Company's Region of employment

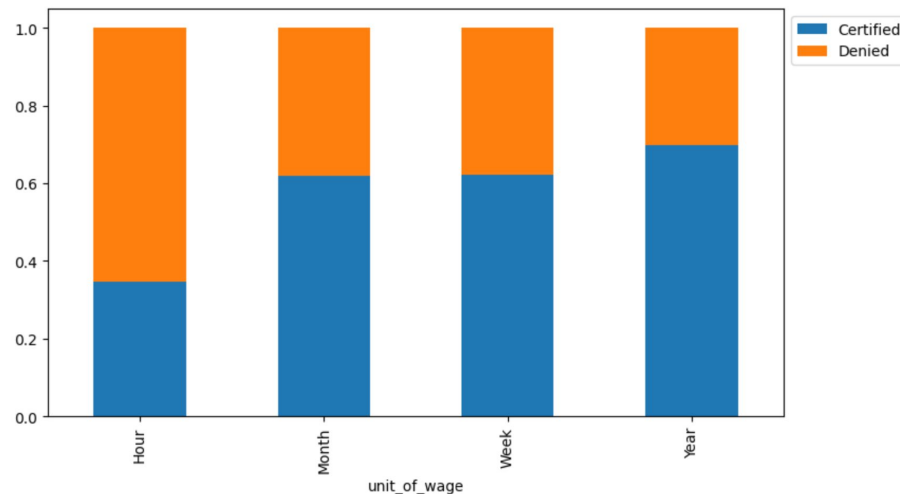


- North east has the highest employment for foreign workers.
- In mainland US, midwest has the least employment for foreign workers.
- Midwest has the higher approvals compared to other regions in US.

EDA : Job and Wage type



- Sixty five percent of the positions are approved in both full and part time positions, indicating an identical approvals.



- Jobs with annual wages are highest approved compared to other wage types.

EDA : Summary

Demographics and Geography

- A significant majority (66%) of all applications originate from Asia.
- Ninety-nine percent of job applications are from the continental U.S. Within the mainland U.S., the Northeast region submits the highest volume of applications, while the Midwest submits the lowest.
- Approval rates vary geographically: The Europe region exhibits the highest visa approval rate overall, whereas South America reports the lowest approval rate across all continents.
- Midwest Approval Anomaly: Despite having the lowest application volume in the U.S., the Midwest region demonstrates higher approval rates compared to other U.S. regions.

Education and Experience

- Applicant education level shows a strong positive correlation with visa approval rates. More than 80% of both doctorate-level and master's-level applications receive approvals.
- Prior job experience is a strong indicator of approval. Applicants who possess relevant job experience consistently receive higher approval ratings than those without experience.

Job Details and Requirements

- Wage Type Matters: Applications listing annual wages have the highest approval rates compared to those with other wage structures (e.g., hourly wages).
- Position Type is Neutral: Approval rates are identical between full-time and part-time positions.
- Training Not a Factor: While a majority of applicants do not require job training, the approval rating is identical for applicants who require training and those who do not.

Data Preprocessing

- **Duplicate check:** No duplicate rows are present in the training data.
- **Missing value check:** Training data has no missing values.
- **Outlier Detection:** The 'number of employees', 'year of establishment', and 'prevailing wage' features have outliers, however given the variability in the data set, these values are expected. Therefore no outliers treatment is performed.
- **Feature Engineering:** is the process to transform the raw data into features that help machine learning algorithm perform better.
 - No new features are created.
 - Data transformed to handle the categorical features.

Data Preprocessing

- **Data preparation for modeling:**

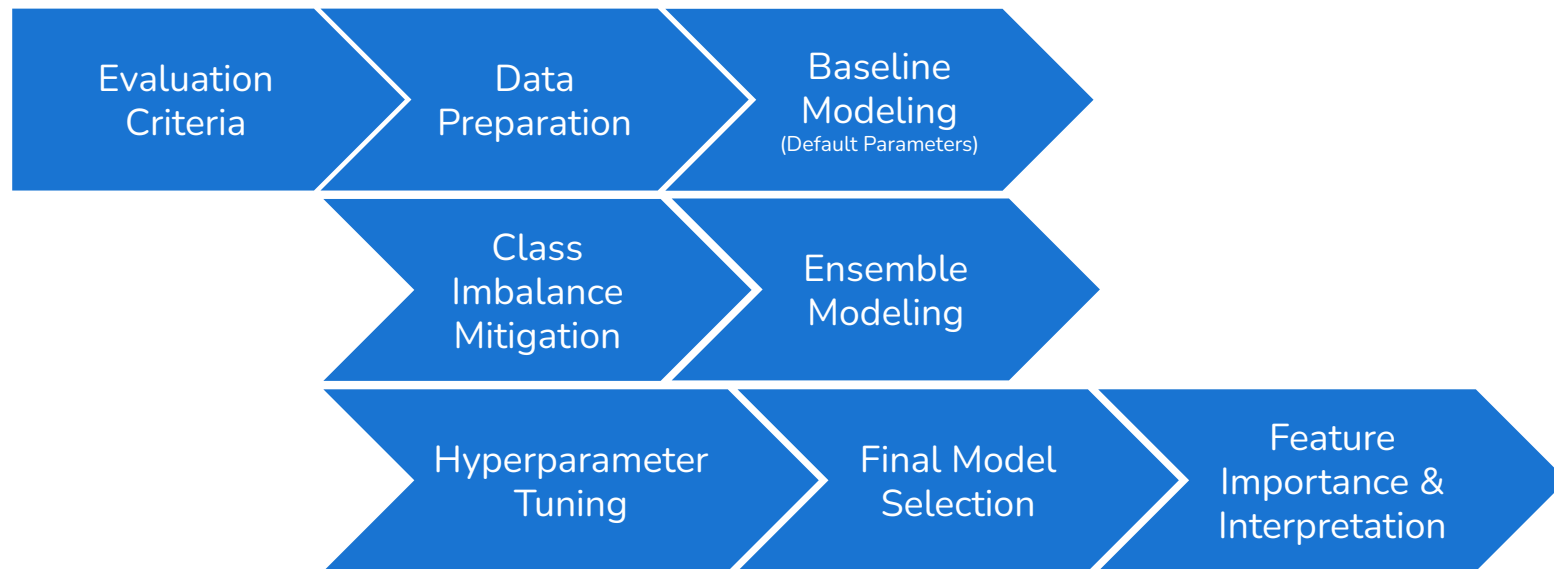
- **Encoding:** Following categorical features are encoded using pandas get dummies function with drop_first option.

- | | |
|-------------------------|------------------------|
| ■ Continent | ■ Region of employment |
| ■ Education of Employee | ■ Unit of wage |
| ■ Has job experience | ■ Full time position |
| ■ Requires job training | |

- **Data Split:**

- Data is split into three sets using stratify- train, validation and test
- Stratify ensures the proportion of target class in each of train, validation and test data sets are maintained.
- train set is used for model training.
- validation set is used for hyper parameter turning to improve the model performance.
- test set is used to evaluate the final model performance.

Methodology: Modeling, Optimization, and Evaluation



Model Evaluation Criteria

Model Predictions:

1. Model predicts that visa application will get certified but in reality, the visa application should get certified.
2. Model predicts that the visa application will get certified but in reality, the visa application should get denied.
3. Model predicts that the visa application will not get certified but in reality, the visa application should get certified.
4. Model predicts that the visa application will not get certified but in reality, the visa application should get denied.



**False
Positive**



**False
Negative**



Important, otherwise
a wrong person will
get the position or
US will lose a
qualified resource

We will use “F1 Score” to evaluate the model performance, a higher score will minimize the chances of False Positives and False Negatives

Model Performance: Base line (Default Parameters)

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9891	0.7736	Overfitting
Random Forest Classifier	1.0	0.8049	Overfitting
AdaBoost Classifier	0.8204	0.8180	Generalizing 
Gradient Boost Classifier	0.8291	0.8266	Generalizing 
XGBoost Classifier	0.8963	0.8079	Overfitting

- AdaBoost and Gradient Boost Classifiers are generalizing well, with a fairly good F1 score of 0.82.
- Bagging, Random Forest and XGBoost classifier models did well on training data, but not so good on validation data indicating a overfit scenario.

Model Performance: Class Oversampling - Default Parameters

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9875	0.7665	Overfitting
Random Forest Classifier	0.9999	0.7965	Overfitting
AdaBoost Classifier	0.8005	0.8195	Generalizing well 
Gradient Boost Classifier	0.8072	0.8173	Generalizing well 
XGBoost Classifier	0.8708	0.8129	Overfitting

- AdaBoost and Gradient Boost Classifiers are generalizing well, with a decent F1 score of 0.81, but it is slightly lower than without Class Oversampling.
- Bagging, Random Forest and XGBoost classifier models did well on training data, but not so good on validation data indicating a overfit scenario.

Model Performance: Class Undersampling - Default Parameters

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9803	0.7057	Overfitting
Random Forest Classifier	1.0	0.7417	Overfitting
AdaBoost Classifier	0.7015	0.76599	Underfitting
Gradient Boost Classifier	0.7281	0.7765	Underfitting
XGBoost Classifier	0.8720	0.7459	Overfitting

- AdaBoost and Gradient Boost Classifiers did better on validation data, but are making lot more mistakes in prediction.
- Bagging, Random Forest and XGBoost classifiers did well on training, but performed poorly on validation data indicating the overfitting.

Tuned Model Performance - Training Data

Training performance comparison:

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.776622	0.749084	0.749804	0.762493
Recall	0.911166	0.876200	0.899119	0.979530
Precision	0.787656	0.776697	0.766597	0.745064
F1	0.844921	0.823454	0.827586	0.846359

- **Random Forest and XGBoost Classifiers** have the best **F1 score** on training data.
- Random Forest has the highest Precision score, ie. lower False Positives
- XGBoost Classifier has the highest Recall score, ie. lower False Negatives.

Tuned Model Performance - Validation Data

Validation performance comparison:

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.749804	0.749411	0.749019	0.725667
Recall	0.890423	0.872503	0.897474	0.946827
Precision	0.770659	0.778914	0.766625	0.725901
F1	0.826223	0.823057	0.826905	0.821775

- **Gradient Boost Classifier** has the best **F1 score** indicating lower False positives and False negatives.
- Also Gradient Boost Classifier is generalizing well with similar F1 scores between Training and Validation data.

Tuned Model Performance - Test Data

Test performance comparison:

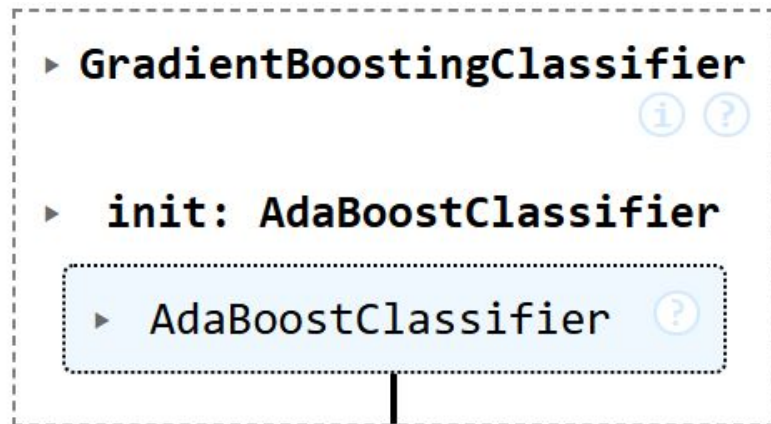
	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.737637	0.739403	0.737441	0.713108
Recall	0.883666	0.873972	0.891892	0.947121
Precision	0.761712	0.767940	0.757863	0.715490
F1	0.818169	0.817532	0.819433	0.815171

- **Gradient Boost Classifier** has the best **F1 score** indicating lower False positives and False negatives.
- Overall tuned **Gradient Boost Classifier** is generalizing well, with similar F1 scores between Training, Validation and Test data.

Model Performance Summary and Final Model Selection

- **AdaBoost Classifier** and **Graident Boost Classifiers** performed well with **default parameters**.
- Class Imbalance techniques such as **SMOTE** oversampling and **RandomUnderSampling** have not resulted in model performance improvements.
- Hyper parameter tuning has improved the overall model performance by reducing the overfitting, ie a similar F1 score is observed across Training, Validation and Test data sets.
- **Tuned Gradient Boost Classifier** and **XGBoost Classifier** have the best F1 scores, however XGBoost is slightly overfitting the training data.
- Based on the observed F1 scores across Training, Validation and Test datasets, We choose **Tuned Gradient Boost Classifier** as the final model.

Final Model



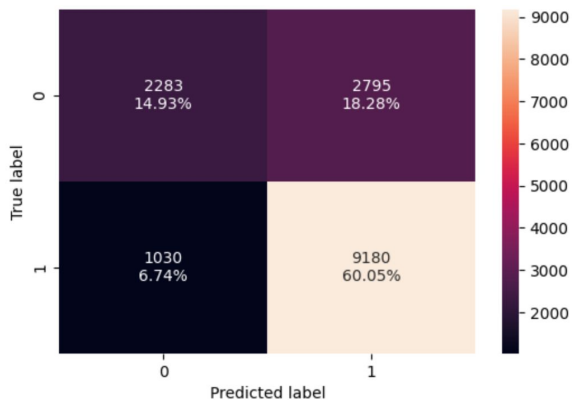
Hyperparameters:

```
"n_estimators": np.arange(50,110,25),  
"subsample": [0.7,0.9],  
"max_features": [0.7, 0.8, 0.9, 1],  
"learning_rate": [0.01,0.1,0.05],
```

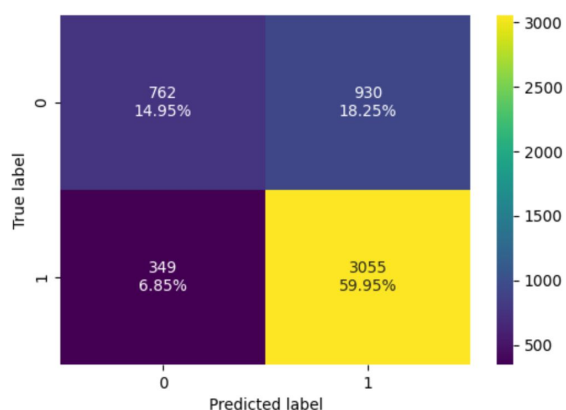
- Model tuning is performed by Grid Search Cross validation with cv=5 folds
- Model is tuned by fitting 5 folds for each of 72 candidates, totalling 360 fits

Final Model : Gradient Boosting Classifier Performance

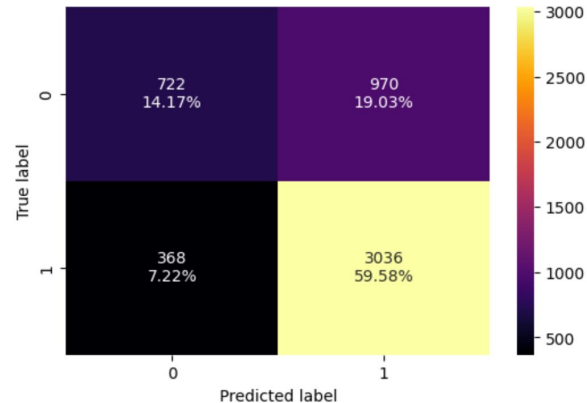
Confusion matrix on Training Data



Confusion matrix on Validation Data

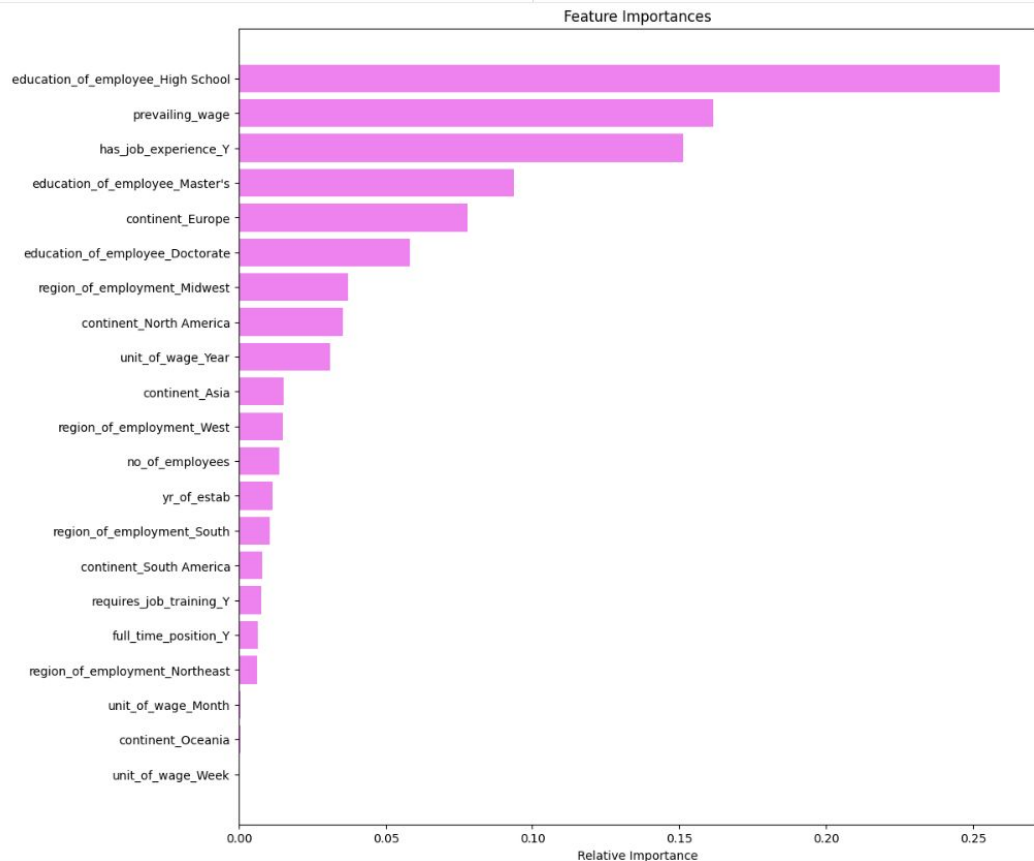


Confusion matrix on Test Data



- ~26% of the applications are predicted incorrectly across the data sets.
- ~7% of applications were False Negatives (Missed approvals)
- ~18% of applications were False Positives (Incorrectly reviewed)
- The performance is identical across the data sets, indicating a fairly well generalized model.
- The model achieves a high Recall (~89%) on the positive class, successfully identifying vast majority of qualifying applications.

Important Features



Primary predictors for Application outcomes:

- **Education level** - Higher education levels show positive correlation with approvals.
- **Prevailing wages** - A positive predictor, higher wages show a strong correlate with approval outcomes
- **Prior Job Experience** - Employees with prior job experience have higher chance of receiving approvals
- **Continent** - European region has higher approvals rate, followed by Africa, Asia, North America and South America regions.

APPENDIX

Data Background and Contents

The data contains the different attributes of employee and the employer. The detailed data dictionary is given below.

- `case_id`: ID of each visa application
- `continent`: Information of continent the employee
- `education_of_employee`: Information of education of the employee
- `has_job_experience`: Does the employee has any job experience? Y= Yes; N = No
- `requires_job_training`: Does the employee require any job training? Y = Yes; N = No
- `no_of_employees`: Number of employees in the employer's company
- `yr_of_estab`: Year in which the employer's company was established
- `region_of_employment`: Information of foreign worker's intended region of employment in the US.
- `prevailing_wage`: Average wage paid to similarly employed workers in a specific occupation in the area of intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- `unit_of_wage`: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- `full_time_position`: Is the position of work full-time? Y = Full Time Position; N = Part Time Position
- `case_status`: Flag indicating if the Visa was certified or denied

Model Performance Summary and Final Model Selection

- Compare the performance of the tuned models
- Overview of final ML model and its parameters
- Comment on the performance of the final model
- Summary of most important factors used by the ML model for prediction

Note: *You can use more than one slide if needed*

Model Building - Original Data

- Five Classifier models with default parameters were developed to establish the baseline performance.

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9891	0.7736	Overfitting
Random Forest Classifier	1.0	0.8049	Overfitting
AdaBoost Classifier	0.8204	0.8180	Generalizing well.
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- AdaBoost and Gradient Boost Classifiers are generalizing well, with a fairly good F1 score of 0.82.
- Bagging, Random Forest and XGBoost classifier models did well on training data, but not so good on validation data indicating a overfit scenario.

Model Building - Oversampled Data

- SMOTE (**S**ynthetic **M**inority **O**versampling **T**echnique) is used to address the imbalance between the output classes. The algorithm uses K-nearest neighbors approach to generate the minority class data.

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9875	0.7665	Overfitting
Random Forest Classifier	0.9999	0.7965	Overfitting
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AdaBoost and Gradient Boost Classifiers are generalizing well, with a decent F1 score of 0.81, but it is slightly lower than without Class Oversampling.

Model Building - Undersampled Data

- RandomUnderSampler method is used to undersample the training data.

Model	F1 Score		Observation
	Training	Validation	
Bagging Classifier	0.9803	0.7057	Overfitting
Random Forest Classifier	1.0	0.7417	Overfitting
AdaBoost Classifier	0.7015	0.76599	Underfitting
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- AdaBoost and Gradient Boost Classifiers did better on validation data, but are making lot more mistakes in prediction.
- Bagging, Random Forest and XGBoost classifiers did well on training, but performed poorly on validation data indicating the overfitting.

Model Improvement - Hyperparameter Tuning

- The goal of Hyperparameter tuning is to find the optimal configuration to maximize the model performance.
- The tuning process should balance between the performance and execution time. Overturning not only takes long execution time, also could lead to overfitting.
- The parameters vary based on the model, the following parameters were tuned.

```
parameters = {
    "max_depth": list(np.arange(5, 15, 5)),
    "max_features": ["sqrt", "log2"],
    "min_samples_split": [3, 5, 7],
    "n_estimators": np.arange(10, 40, 10),
}
```

Random Forest Classifier

```
parameters = {
    # Let's try different max_depth for base_estimator
    "estimator": [
        DecisionTreeClassifier(max_depth=2, random_state=1),
        DecisionTreeClassifier(max_depth=3, random_state=1),
    ],
    "n_estimators": np.arange(50, 110, 25),
    "learning_rate": np.arange(0.01, 0.1, 0.05),
}
```

AdaBoost Classifier

```
parameters = {
    "n_estimators": np.arange(50, 110, 25),
    "subsample": [0.7, 0.9],
    "max_features": [0.7, 0.8, 0.9, 1],
    "learning_rate": [0.01, 0.1, 0.05],
}
```

Gradient Boost Classifier

```
parameters = {
    "n_estimators": np.arange(50, 110, 25),
    "scale_pos_weight": [1, 2, 5],
    "subsample": [0.9, 1],
    "learning_rate": [0.01, 0.1, 0.05],
    "gamma": [1, 3]
}
```

XGBoost Classifier

- Gradient Boost Classifier, XGBoost Classifier and Random Forest Classifier models performed relatively better.
- Overall **Gradient Boost Classifier** has the best F1 score with identical performance between the Training, Validation and Test data indicating good generalization

Model Performance - Training, Validation and Test data

Training performance comparison:

	Tuned Random Forest	Tuned Adaboost Classifier	Tuned Gradient Boost Classifier	XGBoost Classifier Tuned
Accuracy	0.776622	0.749084	0.749804	0.762493
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F1	0.818169	0.817532	0.819433	0.815171

Slide Header

- Please add any other pointers (if needed)



Happy Learning !

